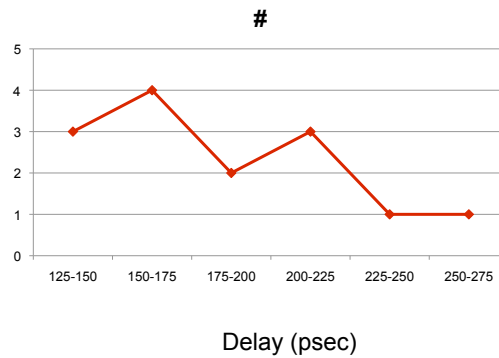


Delay



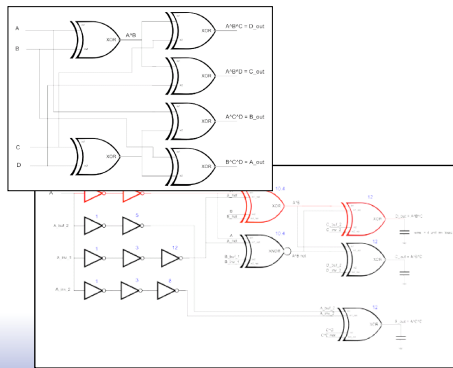
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Example of Good Phase-1 Report

- Starting from Optimal # of Stages
- Correct Hand Calculations
- Optimized Logic Network



C_{load} = bottom plate + fringing + 4 * min sized inverter
 $= 0.25 \text{ mm} * 0.14 \text{ um} * 19.9 \text{ aF/um}^2 + 2 * 0.25 \text{ mm} * 25.6 \text{ aF/um} + 4 * 0.582 \text{ fF}$
 $= 15.82 \text{ fF}$

The total path fanout with our design is (C_{load}/C_{in}) , which would be $(15.82 \text{ fF}/0.582 \text{ fF}) = 27.2$, with a single unit-sized inverter starting the path. The path branch effort is 4 -- 2 for each path that branches to two xor gates. The path logical effort is 4 because each XOR has a logical effort of 2. Therefore, our total path effort is fanout*(branch effort)*(path logical effort) = 434.538.

The optimal number of stages to have a fanout of 3.6 is with $\log(\text{total path effort})/\log(3.6)$, which is approximately 4.7. Since we can only add an even number of inverters, we choose to have 4 stages. With four stages, the stage effort becomes $(434.538)^{(1/4)} = 4.56569$. This implies that a path with 4 stages will have a larger, less optimal fanout, than a path with 5 stages.

We will call the sizes of the path 1, x, y, and z for each of the gates from beginning to end on input A_buf_1.

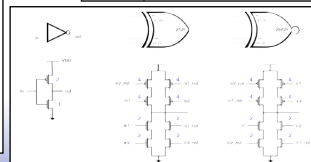
$$f = \frac{15.82 \text{ fF}}{z * C_{in}} = \frac{EF}{LE}$$

$$z = \frac{15.82 \text{ fF} * LE}{EF * C_{in}} = \frac{15.82 \text{ fF} * 2}{4.56 * 0.582 \text{ fF}} = 11.8968 \approx 12$$

$$y = \frac{z * LE_y * B_y}{EF} = \frac{12 * 2 * 2}{4.56} = 10.42 \approx 10.4$$

$$x = 1 * 4.56 = 4.56 \approx 4.5$$

Thus, we get sizes 1, 4.5, 10.4, and 12 for the critical path. The sizes gates on the other paths are sized with these sizes



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Example of Good Phase-1 Report

- ❑ Worst Case Input Pattern Analysis
- ❑ Noise Margin Simulation
- ❑ Well-Organized Report

c) Critical Path: Analytical

1. Identify the critical path(s) and explain the reason.

Show schematics of the check node with highlighted critical path. Show the hand calculation of how you size the gates for minimum delay. Annotate the sizes of all gates along the critical path.

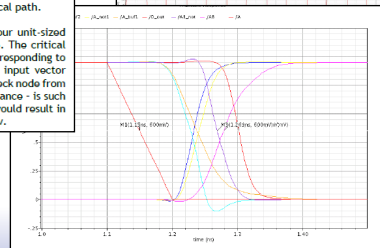
Refer to schematic above with proper sizes and highlighted critical path.

Maximum input capacitance = $4C_{in,inv}$, which we meet with four unit-sized inverters for each branch of the inputs as in the picture above. The critical path is the one with the two inverters followed by two XORs, corresponding to input chain A_buf_1 above. This is for the output D_out, and input vector (A,B,C) = (1 → 0, 1, 1) and (0 → 1, 1, 1). The total fanout of the check node from input to output -- as specified by the max input and load capacitance -- is such that an even number of buffer stages preceding the XOR gates would result in the fanout-per-stage being suboptimal. This calculation is below.

d) Critical Path: Simulation

1. Simulate the functionality of your check node with given test patterns. Show the simulation results that confirm the correctness of your design.

By symmetry of our circuit, only the input for D_out is given below. It is kind of hard to see this pictures, but the graphs are in order of D_out, D, C, B, A. D_out should be A'B'C as seen below. This is the simulation for input swings of 0.1VDD to 0.9VDD, so it is clear that our circuit functions within the noise margins.



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Energy/Power Revisited

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Transition Activity and Power

□ Energy consumed in N cycles, E_N :

$$E_N = C_L \cdot V_{DD}^2 \cdot n_{0 \rightarrow 1}$$

$n_{0 \rightarrow 1}$ – number of $0 \rightarrow 1$ transitions in N cycles

$$P_{avg} = \lim_{N \rightarrow \infty} \frac{E_N}{N} \cdot f = \left(\lim_{N \rightarrow \infty} \frac{n_{0 \rightarrow 1}}{N} \right) \cdot C_L \cdot V_{DD}^2 \cdot f$$

$$\alpha_{0 \rightarrow 1} = \lim_{N \rightarrow \infty} \frac{n_{0 \rightarrow 1}}{N} \cdot f$$

$$P_{avg} = \alpha_{0 \rightarrow 1} \cdot C_L \cdot V_{DD}^2 \cdot f$$

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Factors Affecting Transition Activity

- “Static” component (does not account for timing)
 - ⇒ Type of Logic Function (NOR vs. XOR)
 - ⇒ Type of Logic Style (Static vs. Dynamic)
 - ⇒ Signal Statistics
 - ⇒ Inter-signal Correlations
- “Dynamic” or timing dependent component
 - ⇒ Circuit Topology
 - ⇒ Signal Statistics and Correlations

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Type of Logic Function: NOR

Example: Static 2-input NOR Gate

A	B	Out
0	0	1
0	1	0
1	0	0
1	1	0

Assume **signal probabilities**

$$p_{A=1} = 1/2$$

$$p_{B=1} = 1/2$$

Then **transition probability**

$$p_{0 \rightarrow 1} = p_{Out=0} \times p_{Out=1}$$

$$= 3/4 \times 1/4 = 3/16$$

If inputs switch every cycle

$$\alpha_{0 \rightarrow 1} = 3/16$$

Type of Logic Function: NAND

Example: Static 2-input NAND Gate

A	B	Out
0	0	1
0	1	1
1	0	1
1	1	0

Assume **signal probabilities**

$$p_{A=1} = 1/2$$

$$p_{B=1} = 1/2$$

Then **transition probability**

$$p_{0 \rightarrow 1} = p_{Out=0} \times p_{Out=1}$$

$$= 3/4 \times 1/4 = 3/16$$

If inputs switch every cycle

$$\alpha_{0 \rightarrow 1} = 3/16$$

Type of Logic Function: XOR

Example: Static 2-input XOR Gate

<i>A</i>	<i>B</i>	<i>Out</i>
0	0	0
0	1	1
1	0	1
1	1	0

Assume **signal probabilities**

$$p_{A=1} = 1/2$$

$$p_{B=1} = 1/2$$

Then **transition probability**

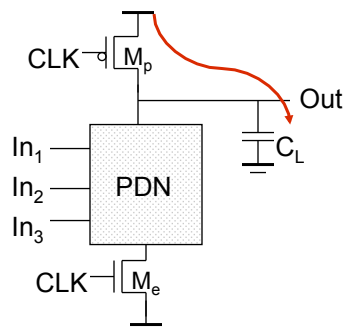
$$p_{0 \rightarrow 1} = p_{Out=0} \times p_{Out=1}$$

=

If inputs switch in every cycle

$$\alpha_{0 \rightarrow 1} =$$

Power Consumption of Dynamic Gates



Power only dissipated when previous Out = 0

Dynamic Power Consumption is Data Dependent

Dynamic 2-input NOR Gate

A	B	Out
0	0	1
0	1	0
1	0	0
1	1	0

Assume **signal probabilities**

$$P_{A=1} = 1/2$$

$$P_{B=1} = 1/2$$

Then **transition probability**

$$P_{0 \rightarrow 1} = P_{\text{out}=0} \times P_{\text{out}=1}$$

$$= 3/4 \times 1 = 3/4$$

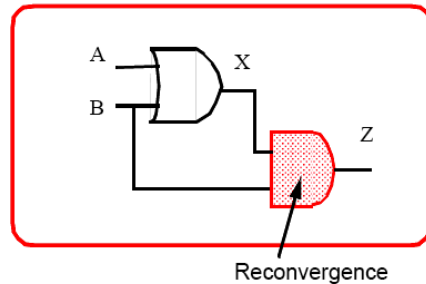
Switching activity always **higher** in dynamic gates!

$$P_{0 \rightarrow 1} = P_{\text{out}=0}$$

Clock

- ❑ Always switches
- ❑ Often consumes 25-50% of total power
- ❑ Clock gating commonly employed

Problem: Reconvergent Fanout



$$P(Z = 1) = P(B = 1) \cdot P(X = 1 \mid B=1)$$

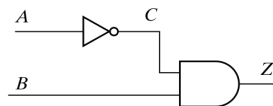
Becomes complex and intractable fast

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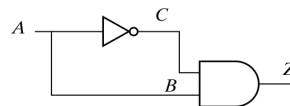
Inter-Signal Correlations



(a) Logic circuit without reconvergent fan-out

Logic without
reconvergent fanout

$$p_{0 \rightarrow 1} = (1 - p_{\bar{A}}p_B) p_{\bar{A}}p_B$$



(b) Logic circuit with reconvergent fan-out

Logic with
reconvergent fanout

$$P(Z = 1) = p(C=1 \mid B=1) p(B=1)$$

$$p_{0 \rightarrow 1} = 0$$

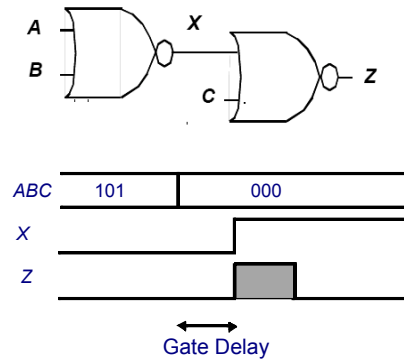
- ❑ Need to use conditional probabilities to model inter-signal correlations
- ❑ CAD tools best for performing such analysis

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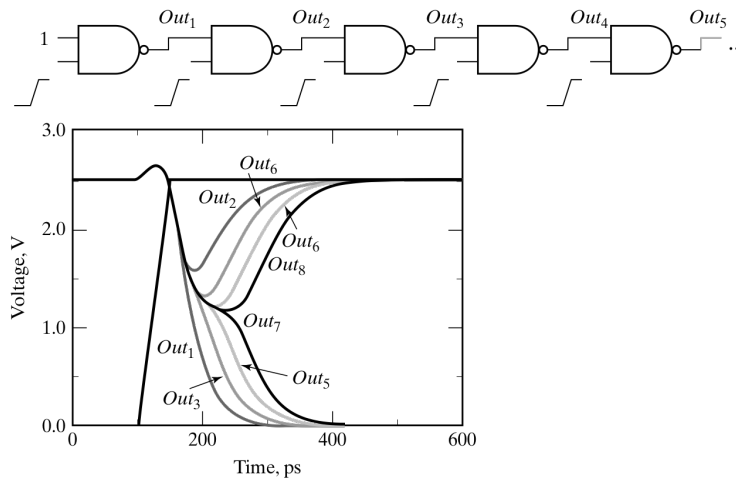
Glitching in Static CMOS



Also known as
dynamic hazards

The result is correct,
but there is extra power dissipated

Example: Chain of NAND Gates



Principles for Power Reduction

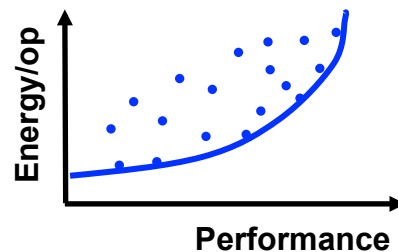
- Most important idea: reduce waste
- Examples:
 - Don't switch capacitors you don't need to
 - Clock gating, glitch elimination, logic re-structuring
 - Don't run circuits faster than needed
 - Power $\propto V_{DD}^2$ – can save a lot by reducing supply for circuits that don't need to be as fast
 - Parallelism falls into this category
- Let's say we do a good job of that – then what?

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Energy – Performance Space



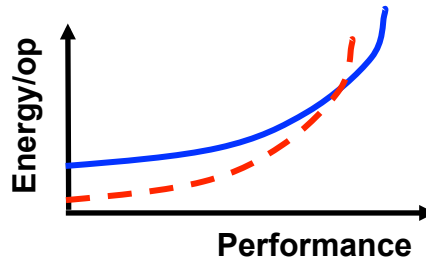
- Plot all possible designs on a 2-D plane
 - No matter what you do, can never get below/to the right of the solid line
- This line is called “Pareto Optimal Curve”
 - Usually (always) follows law of diminishing returns

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Optimization Perspective



- Instead of metrics like EDP, this curve often provides information more directly
 - Ex1: What is minimum energy for XX performance?
 - Ex2: Over what range of performance is a new technique (dotted line) actually beneficial?



Adders